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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

(6) Use of Non-programmable Electronic Pocket Calculator is permissible.

Marks

1. Attempt any FIVE : **10**

(a) State role of solvents in paint.

(b) Define 'Flash Point'.

(c) Name any two 'Emulsifiers'.

(d) List any four additives used for solvent based coatings.

(e) Name any two driers.

(f) Define 'Photo-Initiator'.

(g) Explain role of flame retardant additives.

2. Attempt any THREE : **12**

(a) Classify solvents in detail.

(b) Explain importance of evaporation rate of solvent.

(c) Classify surface active agents.

(d) Explain significance of anti-settling agents.

3. Attempt any THREE : **12**

(a) Explain role of polar and non-polar solvents in coating industries.

(b) Write test procedure for determination of 'Distillation Range'.

(c) Describe significance of biocides in paints.

(d) Explain effect of wax additives on paint properties.

4. Attempt any THREE : **12**

(a) Explain precautions for safe storage and handling on solvents.

(b) Explain co-relation of solvent power of solvent and viscosity of paint.

(c) Explain working mechanism of coalescing agents.

(d) Explain the importance of additives dosage with suitable example.

(e) Explain working mechanism of flow and levelling agents.

5. Attempt any TWO : **12**

(a) Explain properties and applications of following solvents in coating industries with examples;

(i) Alcohols

(ii) Water

(b) Explain properties and applications of following solvents in coating industries with examples;

(i) MTO

(ii) Pine oil

(c) Explain working mechanism of Emulsifier with necessary sketches.

6. Attempt any TWO : **12**

(a) Grinding time of paint is greatly affects by the type and dosage of wetting & dispersing agent, 'Justify'.

(b) Explain properties and applications of corrosion inhibitors.

(c) Explain role of special purpose additives in surface coating industries.